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Fractured Reservoir Simulation

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Abstract

This paper describes the development of a three-dimensional (3D), three-phase model for simulating the flow of water, oil, and gas in a naturally fractured reservoir. A dual porosity system is used to describe the fluids present in the fractures and matrix blocks. Primary flow in the reservoir occurs within the fractures with local exchange of fluids between the fracture system and matrix blocks. The matrix/fracture transfer function is based on an extension of the equation developed by Warren and Root and accounts for capillary pressure, gravity, and viscous forces.

Both the fracture flow equations and matrix/fracture flow are solved implicitly for pressure, water saturation, gas saturation, and saturation pressure.

We present example problems to demonstrate the utility of the model. These include a comparison of our results with previous results: comparisons of individual block matrix/fracture transfers obtained using a detailed 3D grid with results using the fracture model's matrix/fracture transfer function; and 3D field-scale simulations of two- and three-phase flow. The three-phase example illustrates the effect of free gas saturation on oil recovery by waterflooding.

Introduction

Simulation of naturally fractured reservoirs is a challenging task from both a reservoir description and a numerical standpoint. Flow of fluids through the reservoir primarily is through the high-permeability, low-effective-porosity fractures surrounding individual matrix blocks. The matrix blocks contain the majority of the reservoir PV and act as source or sink terms to the fractures. The rate of recovery of oil and gas from a fractured reservoir is a function of several variables, including size and properties of matrix blocks and pressure and saturation history of the fracture system. Ultimate recovery is influenced by block size, wettability, and pressure and saturation history. Specific mechanisms controlling matrix/fracture flow include water/oil imbibition, oil imbibition, gas/oil drainage, and fluid expansion. The study of naturally fractured reservoirs has been the subject of numerous papers over the last four decades. These include laboratory investigations of oil recovery from individual matrix blocks and simulation of single- and multiphase flow in fractured reservoirs.

Warren and Root[1] presented an analytical solution for single-phase, unsteady-state flow in a naturally fractured reservoir and introduced the concept of dual porosity. Their work assumed a continuous uniform fracture system parallel to each of the principal axes of permeability. Superimposed on this system was a set of identical rectangular parallelepipeds representing the matrix blocks.

Mattax and Kyte[2] presented experimental results on water/oil imbibition in laboratory core samples and defined a dimensionless group that relates recovery to time. This work showed that recovery time is proportional to the square root of matrix permeability divided by porosity and is inversely proportional to the square of the characteristic matrix length.

Yamamoto *et al.*[3] developed a compositional model of a single matrix block. Recovery mechanisms for various-size blocks surrounded by oil or gas were studied. Gas/oil capillary pressure hysteresis was includ-

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ed to account for the effect of trapped gas on recovery.

Braester[4] gave an analytical solution for 1D, horizontal flow of two immiscible fluids through a fractured porous medium. Capillary pressure was assumed equal to zero in the fracture and nonzero in the matrix. The effect of pressure drop across a matrix block was included in the analysis.

Previous multiphase, multidimensional simulations of naturally fractured reservoirs include papers by Kazemi *et al.*[5] and Rossen.[6] Kazemi treated two-phase water/oil flow using block Gauss-Seidel to solve iteratively first for fracture pressures and water saturations using last iterate matrix pressures and water saturations. Next matrix pressures and water saturations were calculated using last iterate values of fracture pressures and water saturations. This process was repeated until convergence was reached. No gravity terms were included in this work for matrix/fracture flow and they used the same fracture capillary pressure and upstream relative permeability for both matrix/fracture and fracture flow. Two-dimensional (areal) water/oil examples were included.

Rossen's model used input recovery curves that were a function of pressure, matrix block properties, and the fluid environment of the fracture system. These recovery curves were differentiated with respect to pressure and saturation and then were included in a sequential p/S_w-S_g simulator as semi-implicit source terms. Complete segregation of fluids was assumed in the fracture system. Matrix blocks surrounded by gas were assumed not to transfer water from the matrix to the fracture, and matrix blocks surrounded by water were assumed not to transfer gas from the matrix to the fracture. An example was given simulating gas injection into an oil reservoir with an initial gas cap represented by a 2×25 grid with no communication except in the gas cap.

This paper presents the development of a 3D three-phase, dual-porosity, finite-difference model for simulating naturally fractured reservoirs. The formulation is implicit in pressure, water saturation, and gas saturation or saturation pressure for both matrix/fracture flow and fracture flow. We include provisions for calculating gravity, capillary pressure, and viscous forces for both matrix/fracture and fracture flow, and hysteresis effects on both capillary pressure and relative permeability. We present single-block studies to demonstrate matrix/fracture flow and simulation of this term by the model.

The model can be used to simulate primary depletion, gas injection, and water injection in a naturally fractured reservoir. The degree of fracturing within the reservoir is described by input data and can vary from highly fractured regions in portions of the reservoir to unfractured regions in other areas. Examples are included of 3D field-scale water and gas injection.

Development of Flow Equations

The reservoir is assumed to comprise a continuous fracture system filled with discontinuous matrix blocks. Primary flow in the reservoir occurs within the fractures with local exchange of fluids between the fracture system and matrix blocks. Each block is assumed to have known properties and geometric shape, and all blocks within a given grid block are assumed identical.

Following are the equations describing three-phase, 3D flow in the fracture system written in finite difference form.

Fracture Flow Equations

Water.

$$\Delta[T_w(\Delta p_w-\gamma_w\Delta D)]+\lambda_w(p_{wm}-p_w)-q_w$$
$$=\frac{V_b}{\Delta t}\delta(\phi b_w S_w), \quad \ldots\ldots (1)$$

Oil.

$$\Delta[T_o(\Delta p_o-\gamma_o\Delta D)]+\lambda_o(p_{om}-p_o)-q_o$$
$$=\frac{V_b}{\Delta t}\delta(\phi b_o S_o), \quad \ldots\ldots (2)$$

Gas.

$$\Delta[T_g(\Delta p_g-\gamma_g\Delta D)]+\lambda_g(p_{gm}-p_g)$$
$$+\Delta[T_oR_s(\Delta p_o-\gamma_o\Delta D)]+\lambda_oR_s(p_{om}-p_o)-q_g$$
$$=\frac{V_b}{\Delta t}\delta(\phi b_g S_g+\phi b_o R_s S_o). \quad \ldots\ldots (3)$$

The $\lambda(p_m-p)$ terms represent matrix/fracture fluid exchange and act as source or sink terms in the fracture system. With the exception of the matrix/fracture transfer terms, the fracture flow equations are the same as those normally found in a conventional beta-type simulator.

Finite difference equations describing matrix/fracture flow are as follows.

Matrix/Fracture Flow Equations

Water.

$$-\lambda_w(p_{wm}-p_w)=\frac{V_b}{\Delta t}\delta(\phi b_w S_w)_m. \quad \ldots\ldots (4)$$

Oil.

$$-\lambda_o(p_{om}-p_o)=\frac{V_b}{\Delta t}\delta(\phi b_o S_o)_m. \quad \ldots\ldots (5)$$

Gas.

$$-\lambda_g(p_{gm}-p_g)-\lambda_oR_s(p_{om}-p_o)$$
$$=\frac{V_b}{\Delta t}\delta(\phi b_g S_g+\phi b_o R_s S_o). \quad \ldots\ldots (6)$$

The transmissibilities between the matrix and fracture blocks are calculated as follows for the water equation

$$\lambda_w = 0.001127\sigma k_{rw}\left(\frac{kV_b}{B_w\mu_w}\right)_m. \quad \text{.............(7)}$$

The coefficient σ is a geometric factor that accounts for the surface area of the matrix blocks per unit volume and a characteristic length associated with matrix/fracture flow. This coefficient was introduced first by Warren and Root who presented an equation for σ for single-phase quasi-steady-state flow.

The preceding matrix/fracture flow equations assume 1D, horizontal flow between block centers of the matrix and fracture. To include the effect of gravity in the flow terms, we use pseudorelative permeability and capillary pressure curves[7,8] for both the matrix and fracture. This integration effectively reduces the physical flow from 3D to 2D. Fracture/fracture flow is evaluated by using input fracture capillary pressures, usually set equal to zero.

Eqs. 1 through 6 represent six equations in 12 unknowns. Six additional equations—three for the fracture, and three for the matrix expressing the sum of saturations

$$S_w + S_o + S_g = 1, \quad \text{.........................(8)}$$

and capillary pressure relationships

$$P_{cwo} = p_o - p_w \quad \text{.........................(9)}$$

and

$$P_{cgo} = p_g - p_o, \quad \text{.........................(10)}$$

complete the set of 12 model equations.

Accumulation Terms

The right sides of both the fracture and matrix equations are expanded as follows for the fracture equations.

$$\frac{V_b}{\Delta t}\bar{\delta}(\phi b_w S_w) = C_{11}\delta p + C_{12}\delta S_w + C_{13}\delta S_g$$

$$+ (\phi b_w S_w)^k - (\phi b_w S_w)_n. \quad \text{..................(11)}$$

$$\frac{V_b}{\Delta t}\bar{\delta}(\phi b_o S_o) = C_{21}\delta p + C_{22}\delta S_w + C_{23}\delta S_g$$

$$C_{24}\delta p_s + (\phi b_o S_o)^k - (\phi b_o S_o)_n. \quad \text{.............(12)}$$

$$\frac{V_b}{\Delta t}\bar{\delta}(\phi b_g S_g + \phi b_o R_s S_o) = C_{31}\delta p + C_{32}\delta S_w$$

$$+ C_{33}\delta S_g + C_{34}\delta p_s + (\phi b_g S_g + \phi b_o R_s S_o)^k$$

$$- (\phi b_g S_g + \phi b_o R_s S_o)_n. \quad \text{..................(13)}$$

The C_{ij} coefficients represent partial derivatives of the accumulation terms evaluated at k-level conditions. For example,

$$C_{21} = \frac{V_b}{\Delta t} S_o^k(\phi^k b_o' + b_o^k \phi_i c_f), \quad \text{..............(14)}$$

$$C_{22} = -\frac{V_b}{\Delta t}\phi^k b_o^k, \quad \text{.......................(15)}$$

$$C_{23} = -\frac{V_b}{\Delta t}\phi^k b_o^k, \quad \text{.......................(16)}$$

and

$$C_{24} = \frac{V_b}{\Delta t}\phi^k S_o^k b'_{os}, \quad \text{.....................(17)}$$

where formation volume factor for oil is always calculated as a function of pressure and saturation pressure using the equation

$$b_o = b_{os} + b'_o(p - p_s). \quad \text{....................(18)}$$

For conditions above and below the bubble point, only three unknowns exist: P_1, P_2, and P_3, where P_1 and P_2 are δp and δS_w, respectively, and P_3 is equal to δS_g for saturated oil or δp_s for undersaturated oil. Thus, for saturated cells, δp_s is set equal to δp and the C_{i4} coefficients are added to the C_{i1} coefficients. When a cell is undersaturated, δS_g is set equal to zero and the C_{i3} coefficients are set equal to the C_{i4} values.

Fracture Flow Terms

Fracture flow terms are evaluated implicitly as follows for the oil phase fracture flow term.

$$\Delta[T_o(\Delta p_o - \gamma_o \Delta D)] = \Delta[T_o(\Delta p_o - \gamma_o \Delta D)]^k$$

$$+ \Delta[T_o^k(\Delta\delta p - \Delta\delta P_{cgo})]$$

$$+ (\Delta p - \Delta P_{cgo} - \gamma_o \Delta D)^k \delta T_o, \quad \text{..............(19)}$$

where

$$\delta P_{cgo} = \frac{\partial P_{cgo}}{\partial S_g}\delta S_g, \quad \text{......................(20)}$$

and

$$\delta T_o = \frac{\partial T_o}{\partial S_g}\delta S_g + \frac{\partial T_o}{\partial S_w}\delta S_w, \; p < p_s \quad \text{........(21)}$$

or

$$\delta T_o = \frac{\partial T_o}{\partial p_s}\delta p_s + \frac{\partial T_o}{\partial S_w}\delta S_w, \; p > p_s. \quad \text{........(22)}$$

Relative permeabilities and PVT properties all are evaluated at upstream conditions using latest iterate values of pressure, saturations, and saturation pressure.

Matrix/Fracture Flow Terms

Matrix/fracture flow terms in Eqs. 4, 5, and 6 are calculated implicitly in a manner similar to the fracture flow terms with two exceptions: the $\partial T_o/\partial p_s$ derivatives are not included for flow above the bubble point, and special considerations are given to upstream relative permeabilities when flow is from the fracture to the matrix since flow is governed essentially by matrix properties. Relative permeability to water is limited to matrix k_{rw} evaluated at zero P_{cwo}.

$$k_{rw}=k_{rw}(P_{cwo}=0)S_{wf}. \quad \text{(23)}$$

The term S_{wf} accounts for the fractional coverage of a grid block by water. Oil relative permeability is calculated as

$$k_{ro}=k_{ro}(S_w,S_g)_m S_{of}, \quad \text{(24)}$$

where $k_{ro}(S_w,S_g)_m$ is evaluated using Stone's[9] equation:

$$k_{ro}=(k_{rw}+k_{row})(k_{rg}+k_{rog})-(k_{rw}+k_{rg}). \quad \text{(25)}$$

Gas relative permeability is calculated at a matrix gas saturation of one minus residual oil to gas minus irreducible water saturation,

$$k_{rg}=k_{rg}(1-S_{org}-S_{wc})S_{gf}. \quad \text{(26)}$$

For flow from the matrix to the fracture, matrix saturations are used to evaluate relative permeabilities and capillary pressures from input values.

Hysteresis in both the matrix gas/oil and water/oil relative permeability and capillary pressure is incorporated in the model. The hysteresis model, which includes drainage, imbibition, scanning curves, trapped gas, trapped oil, residual oil, and residual gas, is similar to that proposed by Killough.[10] In this work, however, the scanning functions are assumed reversible.

Trapped-gas saturation is defined as the gas saturation at which the gas/oil capillary pressure becomes zero, and trapped-oil saturation is defined in a similar manner for a water/oil system.

Residual-gas saturations are calculated using the equation developed by Land,[11]

$$S_{gr}=\frac{S_{gH}}{1+CS_{gH}}. \quad \text{(27)}$$

The term S_{gH} is a maximum historical gas saturation.

Drainage, imbibition, and scanning gas/oil capillary pressures are evaluated as a function of gas/oil surface tension[12,13] at block pressure

$$P_{cgo}=\frac{\sigma}{\sigma_I}P_{cgoI}. \quad \text{(28)}$$

The term P_{cgoI} corresponds to input capillary pressure values calculated using surface tension, σ_I.

TABLE 1—DATA FOR SINGLE BLOCK EXAMPLES

	Permeability, md	1
	Matrix porosity, %	0.3
One-foot block		
Grid Dimensions:	$7\times7\times8$	
Grid Spacing*:	$\Delta x=\Delta y=$0.01, 0.1, 0.2, 0.4, 0.2, 0.1, 0.01 ft	
	$\Delta z=$0.01, 0.1, 0.2, 0.2, 0.2, 0.2, 0.1, 0.01 ft	
Ten-foot block		
Grid Dimensions:	$7\times7\times8$	
Grid Spacing*:	$\Delta x=\Delta y=$0.001, 1, 2, 4, 2, 1, 0.001 ft	
	$\Delta z=$0.001, 1, 2, 2, 2, 2, 1, 0.001 ft	

*Fracture values are included.

Rate Calculations

Production rates are calculated semi-implicitly as a function of water and gas saturations and saturation pressure, and, optionally, as a function of pressure and wellbore pressure. For example, oil production from a saturated well producing against a constant bottomhole pressure (BHP) is evaluated as

$$q_{on+1}=q_o^k+\frac{\partial q_o}{\partial p}\delta p+\frac{\partial q_o}{\partial S_w}\delta S_w+\frac{\partial q_o}{\partial S_g}\delta S_g, \quad \text{(29)}$$

where

$$\frac{\partial q_o}{\partial p}=PI_o^k, \quad \text{(30)}$$

$$\frac{\partial q_o}{\partial S_w}=b_o^k q_T \partial f_o/\partial S_w, \quad \text{(31)}$$

and

$$\frac{\partial q_o}{\partial S_g}=b_o^k q_T \partial f_o/\partial S_g. \quad \text{(32)}$$

For wells producing a specified constant oil rate, water and gas rates are evaluated as follows for the water rate

$$q_{wn+1}=q_w^k+\frac{\partial q_w}{\partial S_w}\delta S_w, \quad \text{(33)}$$

where

$$\frac{\partial q_w}{\partial S_w}=\left(\frac{B_o\mu_o q_o}{B_w\mu_w}\right)^k\frac{\partial(k_{rw}/k_{ro})}{\partial S_w}\delta S_w. \quad \text{(34)}$$

An option also is included in the model to perturb rates from time n rather than from k. The formulation is identical to the one preceding except that all quantities evaluated at level k are evaluated at time n, and δ terms such as δp are written as $\bar{\delta} p$.

Solution of Flow Equations

The 12 model equations are reduced to 6 equations in 6 unknowns by substitution of Eqs. 8, 9, and 10 for the matrix and fracture into Eqs. 1 through 6. A further reduction in the number of unknowns can be made by eliminating matrix unknowns in terms of fracture values.

TABLE 2—PVT DATA FOR SINGLE-BLOCK AND 3D EXAMPLES

Pressure (psig)	B_o (RB/STB)	B_g (RB/scf)	R_s (scf/STB)	μ_o (cp)	μ_g (cp)	σ (dyne/cm)
1674.0	1.3001	0.00198	367.0	0.529	0.0162	6.0
2031.0	1.3359	0.00162	447.0	0.487	0.0171	4.7
2530.0	1.3891	0.00130	564.0	0.436	0.0184	3.3
2991.0	1.4425	0.00111	679.0	0.397	0.0197	2.2
3553.0	1.5141	0.000959	832.0	0.351	0.0213	1.28
4110.0	1.5938	0.000855	1000.0	0.310	0.0230	0.72
4544.0	1.6630	0.000795	1143.0	0.278	0.0244	0.444
4935.0	1.7315	0.000751	1285.0	0.248	0.0255	0.255
5255.0	1.7953	0.000720	1413.0	0.229	0.0265	0.155
5545.0	1.8540	0.000696	1530.0	0.210	0.0274	0.090
7000.0	2.1978	0.000600	2259.0	0.109	0.0330	0.050

Original bubble point, psig	5,545
Slope of b_o above p_b, vol/vol-psi	0.000012
Density of stock-tank oil, lbm/cu ft	51.14
Slope of μ_o above p_b, cp/psi	0.0000172
Gas density at standard conditions, lbm/cu ft	0.058
Water formation volume factor, psig	1.07
Water compressibility, vol/vol-psi	$3.5(10^{-6})$
Water viscosity, cp	0.35
Water density at standard conditions, lbm/cu ft	65
Matrix compressibility, vol/vol-psi	$3.5(10^{-6})$
Fracture compressibility, vol/vol-psi	$3.5(10^{-6})$

TABLE 3—RELATIVE PERMEABILITY DATA FOR SINGLE-BLOCK AND 3D EXAMPLES

S_w (%)	k_{rw}	k_{row}	P_{cwo} (psi)
0.20	0.0	1.000	50.0
0.25	0.005	0.860	9.0
0.30	0.010	0.723	2.0
0.35	0.020	0.600	0.5
0.40	0.030	0.492	0.0
0.45	0.045	0.392	−0.4
0.50	0.060	0.304	−1.2
0.60	0.110	0.154	−4.0
0.70	0.180	0.042	−10.0
0.75	0.230	0.0	−40.0

S_g (%)	k_{rg}	k_{rog}	P_{cgo} (psi)
0.0	0.0	1.00	0.075
0.1	0.015	0.70	0.085
0.2	0.050	0.45	0.095
0.3	0.103	0.25	0.115
0.4	0.190	0.11	0.145
0.5	0.310	0.028	0.255
0.55	0.420	0.0	0.386

Elimination of Matrix Unknowns

After expanding the matrix/fracture flow equations in totally implicit form, the resulting equations are written mathematically in matrix form as

$$C\mathbf{P}_m = A\mathbf{P} + \mathbf{B}, \quad \text{(35)}$$

where $\mathbf{P}_m$ is the column vector with elements δp_m, δS_{wm}, and δS_{gm}, and $\mathbf{P}$ is the column vector with elements δp, δS_w, and δS_g. Using Gaussian elimination, $\mathbf{P}_m$ is expressed in terms of $\mathbf{P}$ as

$$\mathbf{P}_m = A'\mathbf{P} + \mathbf{B}'. \quad \text{(36)}$$

The matrix/fracture flow terms, Eqs. 4, 5, and 6, now can be written in terms of fracture unknowns. Substitution of either the right or left sides of Eqs. 4 through 6 into Eqs. 1 through 3 can be made. In this work, the right side accumulation terms were used to allow easy modification of the model for additional transfer mechanisms such as those discussed in the Appendix.

Solution Technique

After reducing the flow equations to three equations and three fracture unknowns, a simultaneous solution for all three unknowns is obtained using the reduced bandwidth direct solution method presented by Price and Coats.[14] Next, matrix unknowns are evaluated using Eq. 36. Following each iteration, saturation checks are made and coefficients are re-evaluated. Convergence criteria are based either on maximum pressure change over the last iteration or on the absolute sum of residuals divided by total production/injection.

Single Matrix Block Studies

Detailed simulations of single matrix blocks surrounded by water or gas are presented in this section to illustrate basic recovery mechanisms and to demonstrate the validity of the postulated matrix/fracture transfer equations. One- and ten-ft (0.3- and 3.0-m) cubic matrix blocks of intermediate wettability and absolute permeability of 1.0 md were modeled. Grid dimensions and matrix properties for these runs are shown in Table 1. PVT data are presented in Table 2 and matrix relative permeability and capillary pressure data are given in Table 3. Gas/oil capillary pressure was calculated as a function of pressure according to the variation of surface tension with pressure.

Water/Oil

Water/oil imbibition above the bubble point was modeled by surrounding the matrix block with water at 6,200 psig (43 MPa). Fracture water/oil capillary pressure was set equal to zero and fracture water relative permeability was set equal to the value corresponding to zero capillary pressure to simulate the boundary condition at the face of the block. Oil recovery vs. time for the 1- and 10-ft (0.3- and 3-m) blocks is shown in Figs. 1 and 2, respectively. Recovery from the 1-ft (0.3-m) block essentially was complete at the end of 10 days, while the imbibition process was still active in the 10-ft (3-m) block after 2 years. Note that the ultimate recovery from the 10-ft (3-m) block is 34.2%, compared with 26% for the 1-ft (0.3-m) block. This additional recovery occurs because of a larger gravity head for the 10-ft (3-m) block. Ultimate recovery corresponds to a water saturation where water/oil capillary pressure equals minus one-half the block height times the difference in water and oil densities.

Calculated recovery using the fracture model with a single cell and values of σ equal to 25 and 0.25 for the 1- and 10-ft (0.3- and 3-m) blocks, respectively, is in excellent agreement with the 3D, single-block results. Warren and Root's equation for σ,

$$\sigma = \frac{4N(N+2)}{L^2}, \quad \text{(37)}$$

gives a value of 60 for the 1-ft (0.3-m) block and 0.6 for the 10-ft (3-m) block. The above difference in σ values is not surprising since Warren and Root's value is for single-phase quasi-steady-state flow and the values shown here are for transient countercurrent flow. Calculating

$$\sigma=\frac{A}{LV_b}, \quad \text{(38)}$$

using area and length values corresponding to the centroid of one of the six equilateral pyramids in a cube of side L, yields

$$\sigma=\frac{36.6}{L^2}. \quad \text{(39)}$$

Thus, values for σ of 36.6 and 0.366 are obtained from this equation for the 1- and 10-ft (0.3- and 3-m) blocks.

Kazemi proposed an equation for σ,

$$\sigma=4\left(\frac{1}{L_x^2}+\frac{1}{L_y^2}+\frac{1}{L_z^2}\right), \quad \text{(40)}$$

which was derived on the basis of a length between matrix and fracture nodes and gives values for σ of 12 for the 1-ft (0.3-m) block and 0.12 for the 10-ft (3-m) block. The single-cell fracture model results were obtained using pseudo water/oil relative permeability and capillary pressure.

Gas/Oil

Oil recovery from matrix blocks with gas in the fractures was modeled using a depletion rate of 0.75 psi/D (5.2 kPa). Initial pressure was set equal to 5,540 psig (38 MPa). Gravity drainage is the primary producing mechanism and occurs when the difference between oil and gas densities times block height is greater than the gas/oil capillary threshold pressure.[15] Ultimate recovery corresponds to a saturation distribution where capillary and gravity forces are equal. Recovery from the 1- and 10-ft (0.3- and 3-m) blocks vs. time is shown in Figs. 3 and 4. Ultimate recovery for the 1-ft (0.3-m) block is 10%, which is obtained in fewer than 60 days, while ultimate recovery for the 10-ft (3-m) block is equal to 46%, which is reached in 2.5 years.

Excellent agreement between recoveries from the single-cell fracture model and the 3D values was obtained using σ values of 2.0 and 0.02 for the 1- and 10-ft (0.3- and 3-m) blocks, respectively. These geometric coefficients were derived assuming 1D vertical flow, which characterizes the gravity drainage process. For this case, L in Eq. 38 is evaluated as one-half of the matrix height and A is equal to the area of the base. The single-cell fracture model results were obtained using gas/oil capillary pseudopressures as a function of both gas saturation and pressure, and pseudo gas/oil relative permeabilities at the initial pressure. Table 4 gives the pseudodata at 5,545 psig (34 MPa).

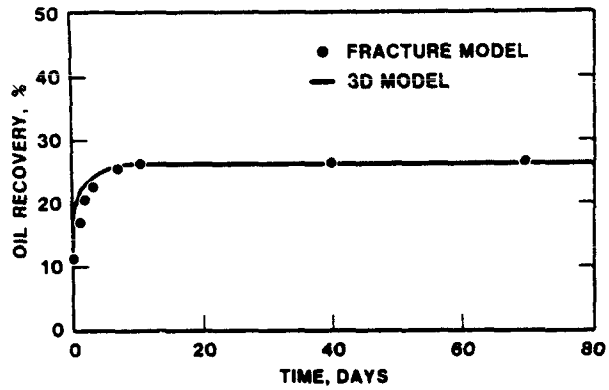

Fig. 1—Water imbibition recovery, 1-ft block.

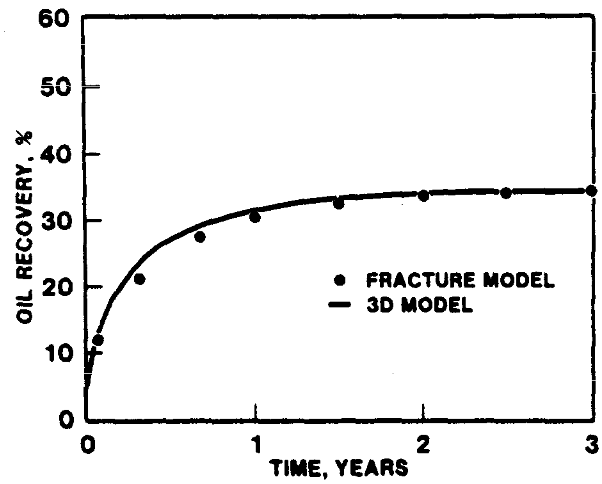

Fig. 2—Water imbibition recovery, 10-ft block.

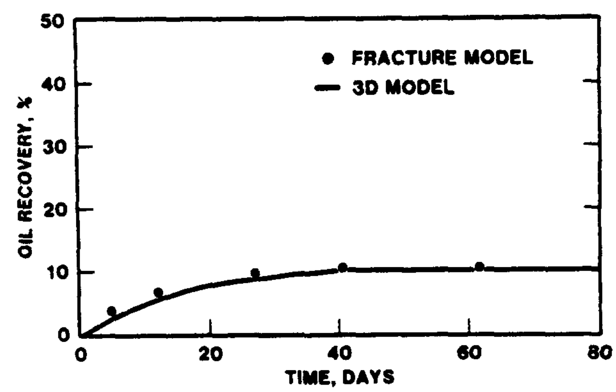

Fig. 3—Gas/oil drainage recovery, 1-ft block.

TABLE 4—GAS/OIL PSEUDODATA AT 5,545 psig

S_g (%)	P_{cgo} (psi)	k_{rg}	k_{ro}
0.0	−0.74	0.0	1.0
0.05	−0.47	0.026	0.89
0.10	−0.33	0.062	0.80
0.20	−0.06	0.14	0.62
0.30	0.21	0.21	0.44
0.40	0.48	0.29	0.25
0.50	0.75	0.37	0.070
0.55	1.27	0.420	0.0

TABLE 5—GAS/OIL IMBIBITION SCANNING CURVES AT 4,500 psig

	$S_{gH}=0.1$		$S_{gH}=0.2$		$S_{gH}=0.3$		$S_{gH}=0.4$	
S_g (%)	k_{rg}	P_{cgo} (psi)	k_{rg}	P_{cgo} (psi)	k_{rg}	P_{cgo} (psi)	k_{rg}	P_{cgo} (psi)
0.076	0.0	−2.0	—	—	—	—	—	—
0.130	0.015	−0.91	0.0	−2.0	—	—	—	—
0.167	0.025	−0.17	0.037	−0.68	0.0	−2.0	—	—
0.193	0.032	0.36	0.045	0.25	0.022	−0.58	0.0	−2.0
0.20	0.034	0.50	0.05	0.50	0.028	−0.20	0.022	−1.04
0.30	0.103	0.57	0.103	0.57	0.103	0.57	0.090	0.018
0.40	0.190	0.73	0.19	0.73	0.19	0.73	0.19	0.73

Three-Phase

Water imbibition below the bubble point into the 10-ft (3-m) block was modeled at 4,500 psig (34 MPa) starting with input gas saturations of 0.4, 0.4, 0.3, 0.2, 0.1, and 0.1 in Layers 1 through 6, respectively, by surrounding the block with water. The gas/oil imbibition scanning curves for each of the above saturations are given in Table 5.

For the single block calculations, residual gas was calculated using a value of C equal to 2.667 in Land's equation. Straight-line scanning curves between maximum gas saturation and residual gas saturation were used for gas/oil relative permeability and capillary pressure.

Fig. 5 shows oil recovery vs. time. Ultimate oil recovery for this example was 47%, which is approximately 13% higher than the recovery from the water/oil example above the bubble point. Fig. 5 also shows a comparison between calculated recovery from the single-cell fracture model and the detailed 3D model and represents an excellent match. A geometric coefficient, σ, of 0.02 was used during the period in which the block was surrounded by gas, and a σ of 0.25 was used after surrounding the block with water. In general, σ is calculated in the model using the gas/oil value, σ_g, if S_{wf} is equal to zero, S_{gf} is greater than zero, and p_{om} is greater than p_{of}. Otherwise, σ is calculated using the water/oil value, σ_w.

Example Problems

Kazemi *et al.* Five-Spot Example

A comparison of results from this paper with previous results is presented in this example which was given first by Kazemi *et al.* Water is injected into one-quarter of a five-spot at a rate of 200 STB/D (31.8 m^3/d) and production is set at a total liquid rate of 210 STB/D (33.4 m^3/d). The reservoir is assumed to be fractured uniformly and was modeled using a 2D grid. Reservoir dimensions and properties are given in Tables 6 and 7. Effective fracture permeability was calculated as

$$K_e=\frac{\phi_f}{\phi_f+\phi_m}K_f. \quad (41)$$

Upstream relative permeabilities were used by Kazemi *et al.* with k_{rw} and k_{row} calculated as a direct function of either fracture or matrix saturation. Capillary pressures given in Table 7 were used for both matrix/fracture flow and flow from one fracture cell to another. Timesteps were calculated automatically using a maximum fracture saturation change of 0.03 and an apparent maximum step size of 20 days. A total of 81 timesteps were required.

Water breakthrough for this example occurred after approximately 30 days, but water production remained relatively low for the first 2 years because of imbibition of water into the matrix rock and countercurrent flow of

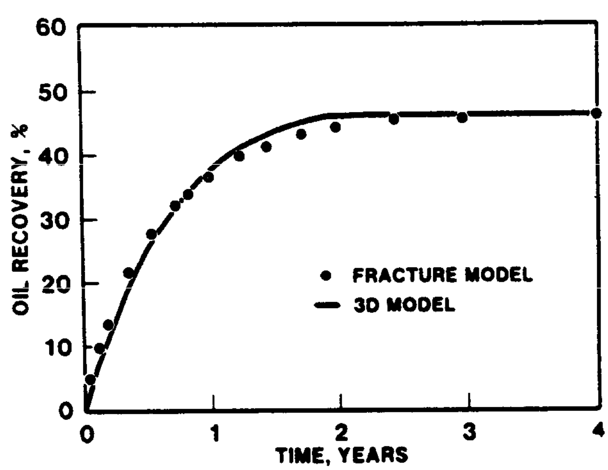

Fig. 4—Gas/oil drainage recovery, 10-ft block.

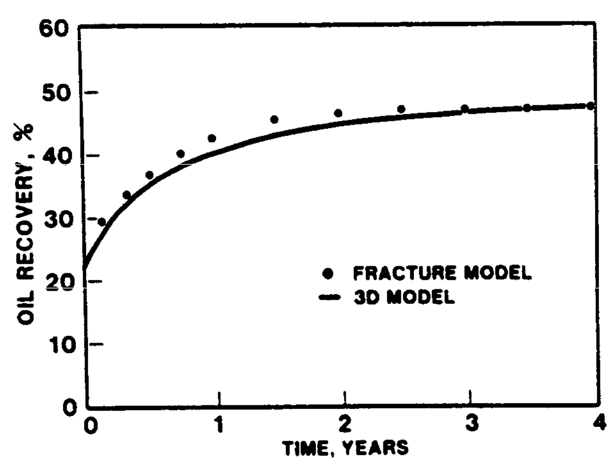

Fig. 5—Three-phase, water imbibition recovery, 10-ft block.

TABLE 6—KAZEMI *et al.* FIVE-SPOT EXAMPLE

Initial pressure, psia	3,959.89
Thickness, ft	30
Grid dimensions	8×8
Grid spacing, $\Delta x = \Delta y$ (ft)	75
Fracture porosity, %	0.01
Matrix porosity, %	0.19
Fracture permeability (effective), md	500
Matrix permeability, md	1
Matrix shape factor, sq ft	0.08
Water compressibility, vol/vol-psi	$3.03 (10^{-6})$
Bubble point pressure, psia	0
Water and oil formation volume factor at the bubble point, RB/STB	1.0
Slope of b_o above p_b, vol/vol-psi	0.0000103093
Fracture compressibility, vol/vol-psi	$3 (10^{-6})$
Water viscosity, cp	0.5
Oil viscosity, cp	2
Water density, psi/ft	0.4444
Oil density, psi/ft	0.3611
Water injection rate, STB/D	200
Total production rate, STB/D	210

TABLE 7—RELATIVE PERMEABILITY DATA FOR KAZEMI *et al.* EXAMPLE

S_w (%)	k_{rwf}	k_{rof}	P_{cwof} (psi)	k_{rwm}	k_{rom}	P_{cwom} (psi)
0.0	0.0	1.000	4.00	—	—	—
0.1	0.05	0.770	1.85	—	—	—
0.2	0.11	0.587	0.90	—	—	—
0.25	0.145	0.519	0.725	0.000	0.920	4.00
0.3	0.180	0.450	0.55	0.020	0.705	2.95
0.4	0.260	0.330	0.40	0.055	0.420	1.65
0.5	0.355	0.240	0.29	0.100	0.240	0.85
0.6	0.475	0.173	0.20	0.145	0.110	0.30
0.7	0.585	0.102	0.16	0.200	0.0	0.00
0.8	0.715	0.057	0.11	—	—	—
0.9	0.850	0.021	0.05	—	—	—
1.0	1.000	0.0	0.0	—	—	—

oil into the fracture system. By the end of 1,200 days, water saturation in the fracture at the injection cell was 0.97 and water saturation in the matrix was 0.58. Similar values were obtained by Kazemi *et al.*

A comparison of water/oil ratio from this work with that calculated by Kazemi *et al.* is shown in Fig. 6 and is quite good considering that we used interpolated values of relative permeability and capillary pressure.

An additional run using a maximum step size of 3 months was made to test the stability of the model. The number of timesteps used in this run was 48 and the oil recovery from the two runs was essentially identical. Both simulations presented here required two iterations per timestep to converge to a pressure tolerance of 0.1 psi (0.7 kPa) and the material balances were 1.0000 for both water and oil at the end of the runs.

Three-Dimensional Examples

Several 3D, field-scale examples are presented to demonstrate the utility of the model and to illustrate the nature of fluid flow in fractured reservoirs. A linear section of reservoir between a production and injection well was modeled in each example. Areally, the reservoir was divided into three cross sections with widths of 800, 400, and 800 ft (244, 122, and 244 m), respectively. Vertically, five layers each having a thickness of 50 ft (15.2 m) were used. In the x direction, the reservoir was divided into ten 200-ft (61-m) grid blocks. Because of the symmetry line through the wells, only one-half of the reservoir was actually modeled using a 10×2×5 grid.

Matrix permeability and porosity were set equal to 1 md and 0.29, and the effective fracture permeability and porosity were 10 md and 0.01, respectively. Matrix

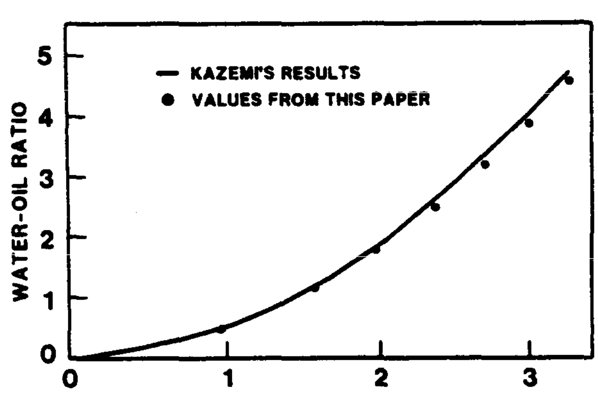

Fig. 6—Kazemi's five-spot example.

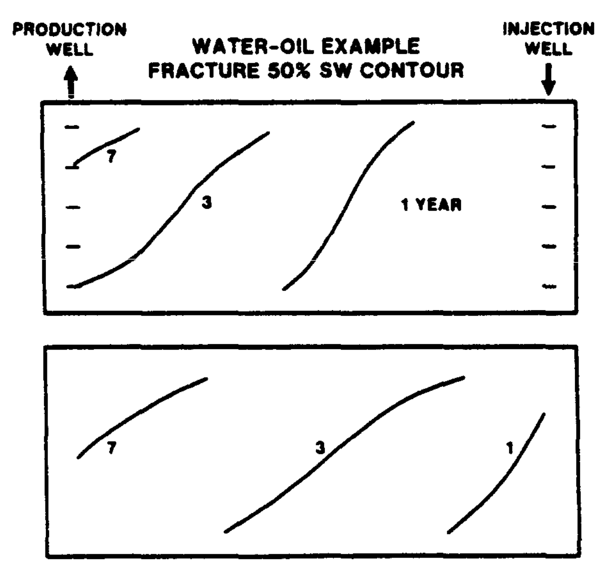

Fig. 7—Water/oil example fracture 50% S_w contour.

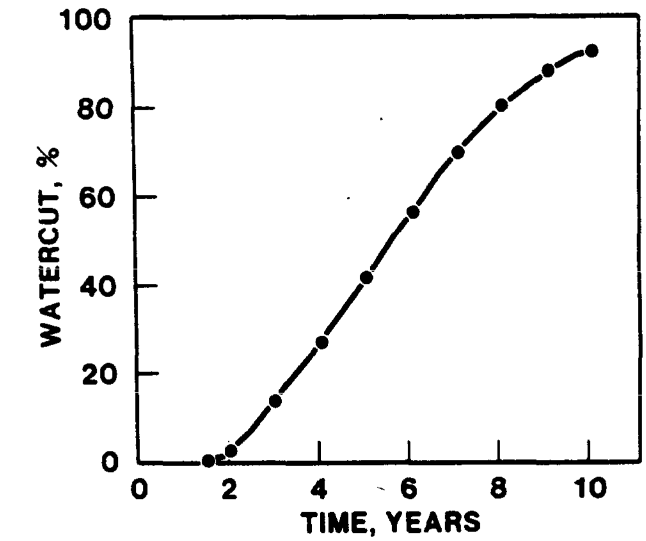

Fig. 8—Water injection example.

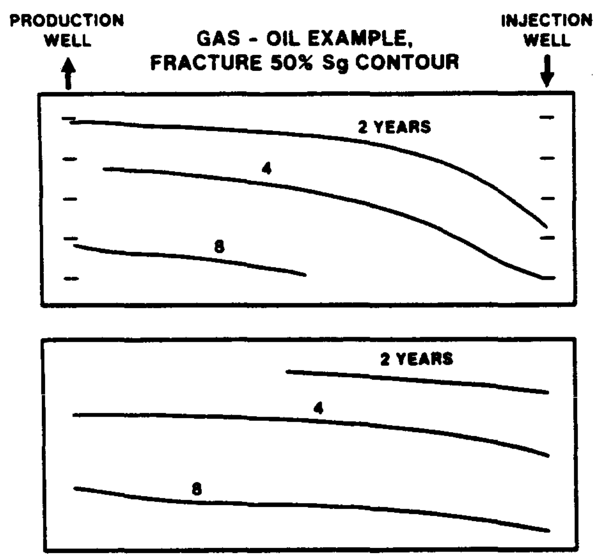

Fig. 9—Gas/oil example, fracture 50% S_g contour.

blocks were assumed to be 10-ft (3-m) cubes. A detailed list of fluid and reservoir properties is given in Table 2.

Initial pressure in the reservoir was set equal to 6,200 psig (43 MPa) at the center of Layer 1. Both wells were perforated in all five layers. Layer productivity indices (excluding $kr/B\mu$) were set equal to 1.0 darcy-ft (0.3 m).

Timesteps were controlled automatically using a maximum saturation change of 0.10, an initial and minimum step size of 0.01 year, and a maximum step size of 0.2 year. Timestep size increase from one step to the next was limited to 1.5 times the previous step size. A pressure tolerance of 0.5 psi (3.45 kPa) was used in all runs.

Water Injection

In this example, water was injected into the reservoir at an initial rate of 7,000 B/D (1113 m^3/d). A maximum flowing bottomhole pressure of 7,400 psig (51.02 MPa) was used to control water injection after pressure buildup near the well. Total liquid rate was set equal to 4,000 STB/D (636 m^3/d).

Water/oil relative permeability and capillary pressure for the matrix is shown in Table 3. We used straight-line fracture relative permeabilities with zero and one as end-points and capillary pressures equal to zero. Rock and water/oil pseudocurves for this example are essentially identical and only rock curves are shown. We used a matrix shape factor equal to 0.25, which was derived from the single block studies for a 10-ft (3-m) block surrounded by water.

Fracture 50% water saturation contours in both the center and outside cross sections at 1, 3, and 7 years are shown in Fig. 7. Note the gravity segregation that occurs in the fracture system, resulting in water underrunning oil. Matrix saturation contours are similar in shape and illustrate the water/oil imbibition process for blocks surrounded by water.

Fig. 8 shows water cut vs. time for this example. Water breakthrough occurred at approximately 1.5 years. Water cut increased gradually, having a value of 56% at 6 years and 92% at 10 years.

Total recovery at the end of 10 years was 35% OOIP. This recovery figure is essentially the same as that obtained in the 10-ft (3-m) water/oil single-block study and indicates that recovery is complete.

This example took 114 timesteps with an average of 2.18 iterations/step. The example was repeated using a maximum saturation change of 0.3 and a maximum timestep size of 0.50 years. This run took 41 steps (3.1 iterations/step) with an average step size of 0.24 years. Water cut at the end of 10 years was 90.5% and oil recovery was within 0.7% of the smaller timestep run. Water and oil material balances at the end of the runs were both 1.0000.

Gas Injection

In this example, 90% of the produced gas was reinjected, resulting in partial pressure maintenance. Injection rates were based on gas production rates at time n. Initial production rate was set equal to 2,000 STB/D (318 m^3/d) and a maximum drawdown of 150 psig (1 MPa) was applied to the well to simulate declining production with increasing GOR. Rock gas/oil relative permeabilities and capillary pressures are given in Table 3. Pseudocurves were input as a function of both saturation and pressure, and two-way interpolation was used to calculate specific values. Data at 5,545 psig (38 MPa) are shown in Table 4. We used a value of σ equal to 0.02 derived from the single-block study for 10-ft (3-m) blocks surrounded by gas.

Injected gas in the fracture system near the well quickly saturated the oil. After the free gas phase was formed, gravity segregation occurred with gas rapidly moving across the top layer of the reservoir. Gas breakthrough resulted after only 0.2 years, and, by the end of 1 year, the gas saturation in the top producing cell was 0.12. Saturation of the matrix blocks took considerably longer, with some of the bottom blocks becoming saturated only after the reservoir pressure was reduced to the original bubble-point pressure of 5,545 psig (38 MPa). Contours of 50% gas saturation at 2, 4, and 8 years for the fracture are given in Fig. 9.

GOR vs. time for this example is shown in Fig. 10. GOR increased sharply after 3 years' operation and reached a value of 118,000 scf/STB (21 254 std m^3/stock-tank m^3) after 9 years. Average pressure in the reservoir at this time was approximately 3,500 psig (24 MPa). Oil recovery after 9 years was 15.4%.

This run took a total of 71 timesteps, averaging 3 iterations/step, corresponding to an average step size of 0.13 years.

Gas and Water Injection

The effect of gas injection for a period of time followed by water injection is demonstrated in this example. The first 4 years' simulation are identical to the gas injection example presented earlier. By the end of 4 years, the GOR in the producing well had risen to approximately 13,000 scf/STB (2342 std m^3/stock-tank m^3), and the average pressure had dropped to 4,700 psig (32 MPa). Starting at the end of the fourth year, the injection well was switched from gas to water injection at a rate of 7,000 B/D (1113 m^3/d). A maximum flowing BHP of 6,000 psig (41 MPa) was employed. At the producing well, a minimum bottomhole flowing pressure of 4,300 psig (30 MPa) and a maximum drawdown of 150 psig (1 MPa) were applied.

Following the initiation of water injection, the GOR at the producing well declined sharply (Fig. 11), until it reached approximately 1,350 scf/STB (243.2 std m^3/stock-tank m^3), corresponding to solution GOR. Water breakthrough in the producing well occurred in approximately 1.5 years.

Residual gas for this example was calculated using Land's equation with a C value of 2.667. Thus, a matrix block with a maximum gas saturation of 0.30 would have a residual gas saturation of approximately 0.17 after water imbibition at constant pressure. Gas saturation contours in the matrix blocks at 10 years are shown in Fig. 12. This figure illustrates the higher trapped gas at the top of the reservoir where significant gravity drainage had occurred. Oil recovery at the end of 16 years was equal to 33.8% and the water cut was equal to 75%.

The 12 years of water flooding in this example took 148 timesteps using a maximum fracture saturation change of 0.1, and 99 timesteps using a maximum fracture saturation change of 0.2. In the latter run, from 6 to 16 years the timestep size was limited by the maximum step size of 0.2 years. The number of iterations per timestep for the two runs was 2.6 and 3.1, respectively. Oil recoveries from the two runs were within 0.3%.

Discussion

It is interesting to compare the recovery from the 10-ft (3-m) single-block run surrounded by gas with the recovery from the 3D field-scale gas injection simulation. The single-block example resulted in an ultimate recovery of 47%, while the 3D field example recovered only 15%. The primary reason for this disparity is that gas segregation to the top of the reservoir allowed only the upper part of the formation to drain oil. Also, gassing zones in the lower part of the reservoir potentially could imbibe oil.

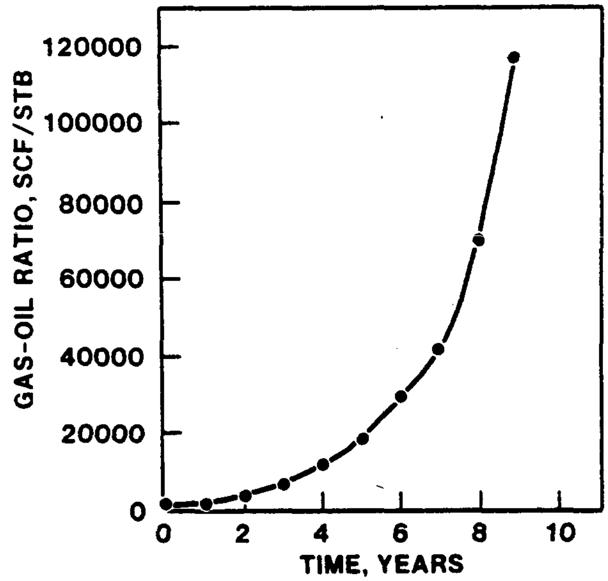

Fig. 10—Gas injection example.

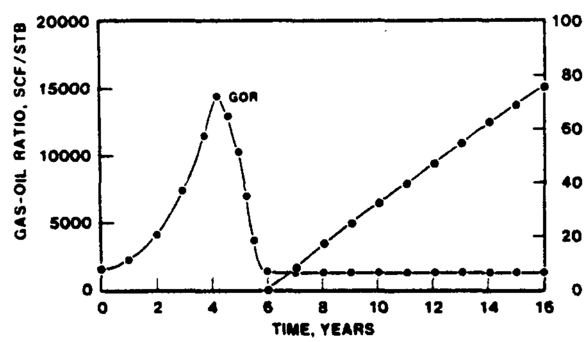

Fig. 11—Three-phase example.

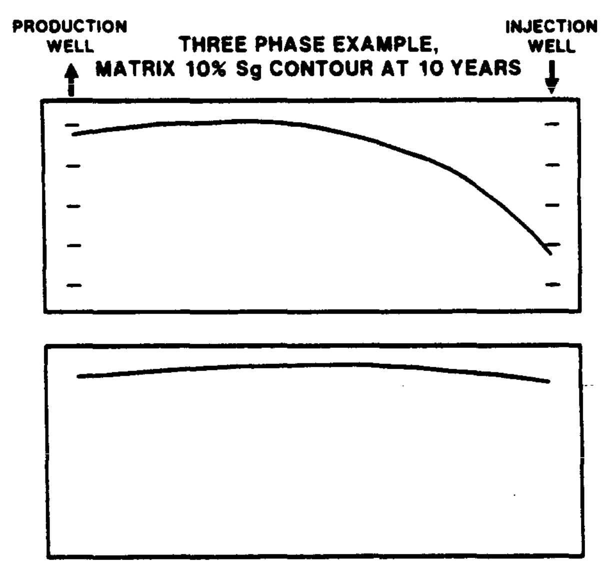

Fig. 12—Three-phase example, matrix 10% S_g contour at 10 years.

To study the effect of better gas coverage, which might be obtained in a multilayered reservoir with limited crossflow, an additional run was made using a vertical/horizontal permeability ratio of 0.1. This run exhibited less gravity segregation than the previous one, but as a result of early gas breakthrough in all layers, recovery was only 10.7% by the time a 100,000 GOR was reached after only after 6 years' operation.

Oil recovery from the water-injection example was essentially the same as the single-block results, indicating that gravity segregation was not a significant problem for this case and that good vertical and areal coverage by water were obtained.

The 3D, three-phase example illustrates the effect of trapped gas on oil recovery. By the end of 12 years' water injection following 4 years' gas injection, a recovery almost equal to ultimate water flood recovery was reached. The water cut at this time was 75% and ultimate recovery was projected to be 37%. Results from the three-phase single-block run indicated 47% recovery. Obviously, this value of recovery can be obtained in a field-scale project only if complete gas coverage is attained prior to waterflooding and if pressure is not increased to a level where the residual gas goes back into solution.

Conclusions

This paper presents the development of a 3D, three-phase model for simulating the flow of fluids in a naturally fractured reservoir. There are four specific advancements in this work.

1. We describe a highly stable formulation that treats both fracture flow and matrix fracture flow implicitly in pressure, water saturation, gas saturation, and saturation pressure.
2. The implicit treatment of matrix/fracture flow presented here can be performed with essentially no additional work per iteration compared to the sequential approach used by Kazemi *et al.* and results in a more stable and efficient model.
3. The matrix/fracture flow equation developed in this paper accurately matches detailed simulations of two-phase, water/oil and gas/oil flow processes as well as three-phase flow. Provisions are included for modeling the gravity term, for properly calculating relative permeabilities as a function of both up- and downstream conditions and hysteresis, and for evaluating the appropriate geometric factor depending on the environment of the fracture system.
4. We give two- and three-phase, 3D examples that demonstrate the utility of the model and provide insight into the nature of multiphase flow in naturally fractured reservoirs.

Nomenclature

A = area, sq ft (m^2)
b = formation volume factor, STB/RB or scf/RB (m^3/m^3)
B = formation volume factor, RB/STB or RB/scf (m^3/m^3)
C = accumulation term partial derivatives
c = compressibility, vol/vol-psi (vol/vol·kPa)
D = depth measured positive downward, ft (m)
f = fractional flow
k = formation permeability, md
k_r = relative permeability
L = length, ft (m)
N = number of normal sets of fractures
p = pressure, psia (kPa)
p_c = capillary pressure, psi (kPa)
$\mathbf{p}_i$ = vector of unknowns
p_s = saturation pressure, psia (kPa)
p_{wf} = flowing bottomhole pressure, psia (kPa)
q = production rate, STB/D or scf/D (std m^3/d)
q_T = total production rate, RB/D (res m^3/d)
q_{mF} = matrix/fracture flow, STB/D or scf/D (std m^3/d)
R_s = solution gas/oil ratio, scf/STB (std m^3/m^3)
S = saturation, fraction
S_{gH} = maximum gas saturation for hysteresis calculation
t = time, days
T = fracture transmissibility, 0.001127 $(kA/L)bk_r/\mu$, STB/D-psi [2.6 (10^{-5}) $(kA/L)bk_r/\mu$, std m^3/d·kPa]
V_b = bulk volume, res bbl (res m^3)
V_p = pore volume, res bbl (res m^3)
W = fracture width, cm (in.)
x,y,z = Cartesian coordinates
γ = specific weight, psi/ft (kPa/m)
δ = iteration difference, $\delta x = x^{k+1} - x^k$
$\bar{\delta}$ = time step difference, $\bar{\delta} = \bar{\delta} x = x_{n+1} + x_n$
Δt = time increment, $t_{n+1} - t_n$
$\Delta(T\Delta p)$ = $\Delta_x(T_x \Delta_x p) + \Delta_y(T_y \Delta_y p) + \Delta_z(T_z \Delta_z p)$
$\Delta(T_x \Delta_x p)$ = $T_{i+½}(p_{i+1} - p_i) - T_{i-½}(p_i - p_{i-1})$
λ = matrix/fracture transmissibility, STB/D-psi (m^3/d·kPa)
μ = viscosity, cp (Pa·s)
ρ = density, lbm/cu ft (kg/m^3)
σ = matrix shape factor, 1/sq ft (1/m^2) or surface tension, dyne/cm (mN/m)
ϕ = porosity
ϕ_i = porosity at initial conditions, fraction

Superscripts

k = iteration level
$'$ = derivative at kth iteration level

Subscripts

e = effective
f = fracture or formation
g = gas

i, j, k = grid block indices
m = matrix
n = time step level
o = oil
r = residual
s = saturation pressure
w = water

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APPENDIX
Semi-Implicit Formulation

A semi-implicit formulation, which uses the implicit matrix/fracture flow described in the text and an implicit fracture solution for pressure and gas saturation or saturation pressure followed by a sequential water calculation, also is included in the model. This formulation is adequate for many two-phase water/oil problems and can be used for some three-phase studies. This formulation offers definite advantages over a strictly sequential approach, especially for conventional nonfractured reservoir simulation.[16]

Matrix/Fracture Transfer

In addition to flow contributions from capillary pressure, gravity, and viscous forces within matrix blocks, matrix/fracture flow may result from a pressure gradient across the matrix block and from diffusion of gas from a saturated fracture cell to an undersaturated matrix block during gas injection. Inclusion of these terms in the matrix/fracture flow equations results in the following equations.

$$q_{wmf}=\lambda_w(p_{wm}-p_w)+\lambda_w\frac{L_c}{L_B}\Delta p_{wf}, \quad \ldots\ldots (A\text{-}1)$$

$$q_{omf}=\lambda_o(p_{om}-p_o)+\lambda_o\frac{L_c}{L_B}\Delta p_{of}, \quad \ldots\ldots (A\text{-}2)$$

and

$$q_{gmf}=\lambda_g(p_{gm}-p_g)+\lambda_g\frac{L_c}{L_B}\Delta p_{gf}$$

$$+\lambda_o R_s(p_{om}-p_o)+\lambda_o R_s\frac{L_c}{L_B}\Delta p_{of}$$

$$+\lambda_{gD}(b_o R_{sm}-b_o R_{sf}, \quad \ldots\ldots (A\text{-}3)$$

where

Δp = pressure drop across matrix block,
L_B = distance over which Δp acts,
L_c = characteristic length for matrix/fracture flow,
λ_{gD} = $\sigma V_b S_{gf}\, D/5.6146$, B/D,
D = diffusion coefficient.

Calculation of Fracture Properties

In-situ fracture permeabilities are a function of fracture width squared.[17] Expressing W in centimeters and k_f in millidarcies yields

$$k_f=8.44\,(10^9)\,W^2. \quad \ldots\ldots (A\text{-}4)$$

Effective fracture permeabilities can be derived[18] by assuming linear flow through the fracture system parallel to four faces of a cubic matrix block of length L. The expression for effective permeability for this system can be written as

$$K_e=\frac{K_m A_m+K_f A_f}{A}. \quad \ldots\ldots (A\text{-}5)$$

The term A represents the area included by one face of the matrix block plus the minor area corresponding to one-half fracture width surrounding the matrix block.

For $W << L$,

$$\frac{A_f}{A}=\frac{2W}{L} \quad \text{(A-6)}$$

and

$$K_e=K_m+K_f A_f/A$$

$$=K_m+1.69\,(10^{10})\frac{W^3}{L}. \quad \text{(A-7)}$$

Fracture porosity for this system is equal to

$$\phi_f=\frac{3W}{L}. \quad \text{(A-8)}$$

SI Metric Conversion Factors

bbl	× 1.589 873	E−01	=	m^3
cp	× 1.0*	E−03	=	Pa·s
cu ft	× 2.831 685	E−02	=	m^3
dyne	× 1.0*	E−02	=	mN
ft	× 3.048*	E−01	=	m
in.	× 2.54*	E+00	=	cm
lbm	× 4.535 924	E−01	=	kg
psi	× 6.894 757	E+00	=	kPa
scf	× 2.863 640	E−02	=	std m^3

*Conversion factor is exact.

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